FIBER OPTIC CABLE – OTDR SUMMARY

Location	Site-A to Site B
Distance	4.2km
Inspection Date	13 August 2026
Cable Specification	48-Core Single-Mode Fiber

Loss Budget Calculation

Loss Budget = [fiber length(km) x fiber attenuation/km] + [splice loss x #of splice] +

[connector loss x #of connectors]

1310nm = [4.20 km x 0.4dB] + [0.2 x 2] + [0.5x 2] = **3.08 dB**

1550nm = [4.20km x 0.3dB] + [0.2x 2] + [0.5x 2] = **2.66 dB**

*Acceptable Loss = Total Loss < Loss Budget

OTDR Test Results Summary

Core	Wavelength 1310nm			Wavelength 1550nm			
	Distance(km)	Total Loss(dB)	Avg Loss(dB)	Distance(km)	Total Loss(dB)	Avg Loss(dB)	Status
Core 1	4.20	2.15	0.51	4.20	1.82	0.43	PASS
Core 2	4.20	2.20	0.52	4.20	1.85	0.44	PASS
Core 3	4.20	2.10	0.50	4.20	1.78	0.42	PASS
Core 4	4.20	2.25	0.53	4.20	1.88	0.45	PASS
Core 5	4.20	2.18	0.52	4.20	1.80	0.43	PASS
***	***	***	***	***	***	***	***
Core 48	4.20	2.22	0.53	4.20	1.84	0.44	PASS